A 19 fJ/op, Low-Offset StrongARM Latch Comparator for Low-Power High-Speed Applications

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***Abstract*—In this paper, a new StrongARM latch comparator design has been proposed for low-power high-speed applications. The proposed design improves the energy consumption and propagation delay when compared to the previous designs in the literature. The proposed design is post-layout simulated in TSMC 65 nm technology node and it achieves a low energy consumption of 19.31 fJ per operation and a low propagation delay of 211 ps. Moreover, the proposed design shows a highly favorable input offset voltage of 0.56 mV and achieves a maximum frequency of 8 GHz. Furthermore, the proposed design reduced the transistor stack that allows it to be used in the low-voltage supply application.**

Keywords—StrongARM latch comparator, low-power circuit, high-speed circuit, offset voltage, CMOS latch, forward body biasing.

I. Introduction

The application of portable electronic systems such as battery-powered biomedical devices increases the requirement for low-voltage and low-power circuit techniques. The StrongARM latch comparator is an essential building block used in the analog-to-digital (ADC) converter which interfaces the biomedical sensor with the digital world [1]. The basic structure of the StrongARM latch comparator was originally introduced by Toshiba's Kobayashi et al. [2] then it was improved multiple times [3-6]. The comparator was named as "StrongARM" after being used in the Digital Equipment Corporation's StrongARM microprocessor [7]. The StrongARM latch comparator is used in most recent interface circuits for its zero static power consumption, produces rail-to-rail output, and its input-referred offset arises from primarily one differential pair. The main design parameters for the StrongARM comparator are speed, power consumption, input offset voltage, input referred noise and kickback noise [8].

This paper presents a novel comparator design that offers low offset voltage, low power consumption, and high speed. The design addresses the need for advanced and scaled-down electronics, particularly in analog-to-digital converters (ADCs), where low-power comparators with accurate effective number of bits (ENOB) are crucial. By reducing power consumption, minimizing offset voltage, and ensuring high-speed operation, the proposed design aims to enhance the performance and efficiency of ADC systems.

II. Standard StrongARM Latch Comparator Design

Typically, the StrongARM latch comparator in Fig. 1 (a) consists of three phases: reset, amplification, and regeneration phases. Reset phase starts when the clock goes low and the nodes

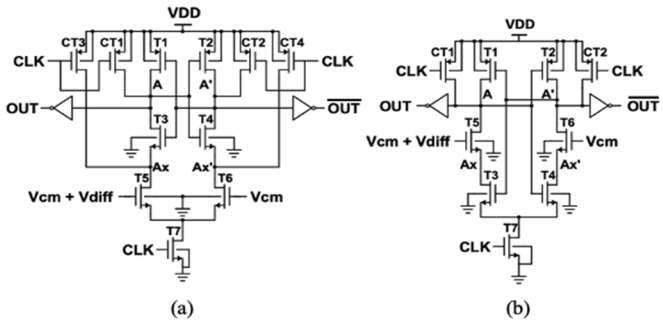


Fig. 1. Schematic of the StrongARM latch comparators: (a) Standard Design [3], (b) Improved Design [4].

A, A', Ax and Ax' charge to V_{DD} through the charging transistors (CT1, CT2, CT3, and CT4). When the clock goes high, the amplification phase starts by turning all charging transistors OFF and turning the tail transistor (T7) ON to allow the node A or A' to discharge through T7. A small voltage variance (V_{diff}) with the common mode voltage (V_{CM}) across the input transistors (T5 and T6) will cause nodes A and A' to discharge at different rates, once the voltage across A or A' nodes reach V_{DD} - V_{th}, the regeneration phase starts where T1 or T2 will start conducting to rebuild a high potential in one of the nodes based on the polarity of the differential input voltage (V_{diff}).

III. Proposed StrongARM Latch Comparator Design

The standard design (Fig. 1 (a)) uses 4 charging transistors (CT1 – CT4) to charge the nodes (A, A', Ax, and Ax') and uses a clocked tail transistor (T7) to discharge those nodes. The standard design was improved by replacing the input transistors (T5 and T6) between the cross-couple transistors (T1 – T4), as shown in Fig. 1 (b), to eliminate (CT3 and CT4) and reduce the total internal capacitance in the circuit. In the proposed design (Fig. 2), the input transistors (T5 and T6) are used to switch the comparator ON and OFF, consequently, the tail transistor (T7) in the standard designs was removed, as a result, the transistors stack is reduced, which allow us to reduce the voltage supply to make the comparator consume less power. The $\overline{CLK}$ is used as a select signal to provide the input transistors with either input signal or ground. During the reset phase, both A and A' nodes are charged to V_{DD} through CT1 and CT2. This will turn ON T3 and T4, which will connect the nodes Ax and Ax' to ground. Contrasting with the standard design, the proposed design turns T5 and T6 OFF to disconnect the upper and lower halves of the comparator. By doing this, nodes A and A'

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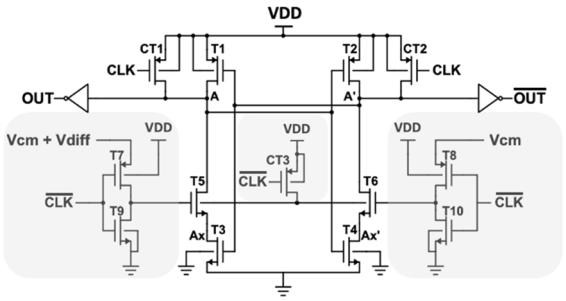


Fig. 2. The proposed StrongARM latch comparator.

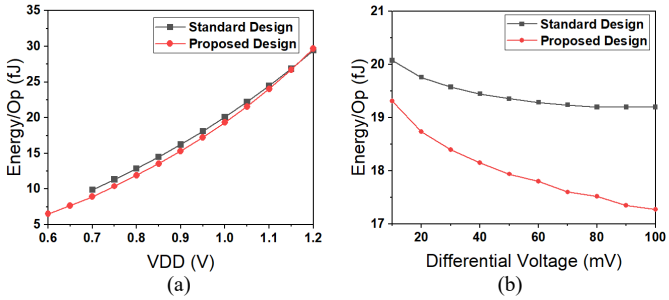


Fig. 3. Energy consumption of the proposed and standard designs: (a) across the voltage supply, (b) across the differential voltage.

will be charged to strong V_{DD} and nodes Ax and Ax' will be discharged to ground. Therefore, once the evaluation phase starts, a higher current pass from drain to source of the input transistors, thus, the energy consumption and the propagation delay are improved. Moreover, the input transistors gate-source voltage (V_{GS}) of the proposed design is higher than the standard design because of the low potential at the source terminals, hence, more current (I_d) is flowing. As the transconductance of the input differential transistors is a function of both I_d and V_{GS}, the amplification gain is higher and the divergence of A and A' signals occur earlier as a consequence.

In order to guarantee proper operation of the proposed design, the input transistors (T5 and T6) are biased with the common mode voltage (V_{CM}) which can be calculated as follows:

$$V_{CM} \geq |V_{DSsat(T7,T8)}| + V_{GS(T5,T6)} + V_{DSsat(T3,T4)} \quad (1)$$

where V_{DSsat} is the drain-source voltage at saturation region and V_{GS} is the gate-source voltage. Since the proposed design is a clocked comparator, the high and low values of the clock signal ($V_{CLK\text{-}low}$ and $V_{CLK\text{-}high}$) can be determined as follows:

$$V_{CLK\text{-}low} \leq V_{CM} - |V_{GS(T7,T8)}| \quad (2)$$

$$V_{CLK\text{-}high} \geq V_{GS(T9,T10)} \quad (3)$$

To improve the propagation delay further, a clock-driven forward body biasing (CFBB) PMOS transistor is used to connect the body terminals of the input transistors (T5 and T6) to V_{DD}, by doing this, the source-body voltage (V_{SB}) is reduced, as a result, the threshold voltage also reduced and makes the input transistors responding faster. When the $\overline{CLK}$ goes low, the CFBB transistor turns ON and charges the bodies of T5 and T6 to V_{DD} in order to lower the threshold voltage and make the input transistors switch ON faster. When the $\overline{CLK}$ goes high, the CFBB transistor turns OFF and a capacitive voltage divider will occur in the internal capacitance of T5, T6 and CT3 and increase the threshold voltage of the input transistors to reduce the leakage current. The sizing of T5, T6 and CT3 is chosen carefully to obtain the optimum body voltage at the regeneration phase. The employment of the CFBB technique requires a triple-well CMOS process to prevent the transistors from latching up, and this process is widely adopted in all recent deep CMOS technologies.

IV. Results and Discussion

The proposed comparator was post-layout simulated using Cadence Virtuoso simulation tools with TSMC 65 nm Process Design Kit (PDK). The power supply (V_{DD}) was set as 1 V and the clock frequency as 250 MHz for all simulations. The energy consumption and propagation delay were analyzed and discussed in this section.

A. Energy Consumption

The tail transistor in the standard design was removed in the proposed design, which reduces the stack transistors, hence, the supply voltage headroom voltage can be reduced, this allows the proposed design to work at a low supply voltage down to 0.6V and reduce its energy consumption. The internal capacitance C_A, $C_{A'}$, C_{Ax}, and $C_{Ax'}$ in Fig. 2 play a big role in the energy consumption, assuming $C_A = C_{A'}$ and $C_{AX} = C_{AX'}$, the circuit's energy consumption in the signal path is given by (4):

$$Energy\ Consumption = f_{CK}\, C_{A,A'}\, V_{DD}^2 + f_{CK}\, 2C_{Ax,Ax'}\, V_{DD}^2 \quad (4)$$

where f_{CK} is the clock frequency and the factor of 2 accounts for the discharge of both Ax and Ax' near ground in every cycle. Additionally, the clock path draws an energy consumption and it can be calculated as shown in (5):

$$Clock\ Energy\ Consumption = f_{CK}\, C_{CK}\, V_{DD}^2 \quad (5)$$

where C_{CK} is the total gate capacitance of the charging transistors (CT1, CT2, and CT3) and the switching transistors (T7, T8, T9, and T10).

Fig. 3 (a) shows the energy consumption per operation for the proposed and the standard designs across the voltage supply range. It can be easily observed that energy consumption is reduced when the voltage supply is reduced. The energy consumption of the proposed design at 0.6 V was 6.5 fJ and at 1.2 V was 29.7 fJ. The standard design, which comprises a stack of four transistors, exhibited a failure to operate below 0.7V.

The input differential voltage affects the comparator behavior, higher input differential voltage offers better performance. The differential voltage was swept from 10 mV to 100 mV and the energy consumption was observed as shown in Fig. 3 (b), as expected, both designs consume less energy at higher input differential voltage and the proposed design showed a better result compared with the standard design.

B. Propagation Delay

The comparator propagation delay is one of the main concerns when designing a comparator for high-speed applications. The propagation delay of the proposed design was improved compared to the standard designs. Fig. 4 (a) shows the propagation

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delay across the voltage supply, when the voltage supply goes below 0.8 V, the comparator becomes exponentially slower and it will stop operating when the overhead voltage is not enough to supply the transistors stack.

The input differential voltage affects the propagation delay as mentioned previously, the input differential voltage was swept from 10 mV to 100 mV as shown in Fig. 4 (b), and the propagation delay was recorded. When the input differential voltage increases, a higher current will flow, and the node will be discharged faster, therefore, the propagation delay will improve.

C. Load Capacitance Impact

The comparator output is always connected to a load, it can be another comparator, shift register, encoder or output pad. The proposed design was tested with different capacitance load values, the load capacitance was swept from 1 fF to 10 fF and the energy consumption and propagation delay were observed. As you can see in Fig. 5, the energy consumption and propagation delay of both designs are affected by the load capacitance, more capacitance, slower performance and higher energy consumption.

D. Input Offset Voltage

The input offset voltage is an important element when designing a comparator especially when the comparator is used in the ADC, a higher offset voltage leads to more errors. There are some methods for lowering the offset voltage [11, 12], however, using such methods will slow down the performance and increase the chip area. The presence of input offset voltage is often due to the mismatch in the transistor's threshold voltage (ΔV_t), current factor ($\Delta ß$), and internal capacitance. The input transistors are the most critical transistors affected by the mismatch. The proposed design introduces a novel technique where it can pull down the input transistor source terminals (T5 and T6) to zero and pull up its drain terminals to V_{DD} at the reset phase, hence, the mismatch of the design is reduced. To validate the reduction in input offset voltage of the proposed design, Monte Carlo simulations were conducted to compare the performance of the proposed design with three other designs. As depicted in Fig. 6 (d), the proposed design exhibits a notably superior input offset voltage in comparison to other designs. The simulation was conducted for a total of 200 iterations and the mean value of the offset voltage was found to be 1.23 mV, with a standard deviation of 0.56 mV for the proposed design. The closest standard deviation from other designs was determined to be 28.62 mV.

E. PVT Corners Variations Simulation

To guarantee our design is functioning well after fabrication, the design was simulated at different process, voltage and temperature (PVT) corners. Figs. 7 (a) and (b) show the effect of PVT variations on the propagation delay, noticeably, the best propagation delay occurs when the NMOS and PMOS operate at the fastest condition (FF), conversely, the worse propagation delay occurs when the transistors operate at the slowest condition (SS). The comparator behaves normally when the transistors are at the fast-slow (FS), slow-fast (SF) and typical-typical (TT) conditions. The same simulation was performed for the energy consumption as illustrated in Figs. 7(c) and (d). As expected, energy consumption is highly affected by the temperature and voltage supply variations, the comparator consumes high energy when NMOS and PMOS are ON, which

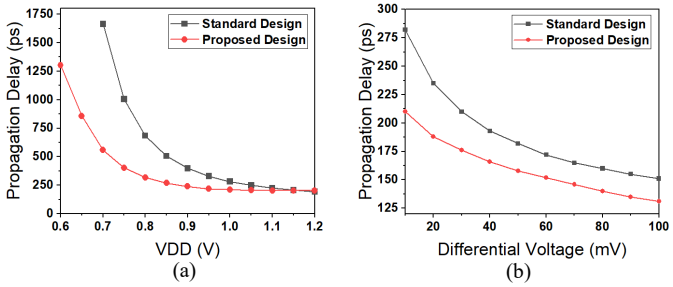


Fig. 4. Propagation delay of the proposed design: (a) across the voltage supply, (b) across the differential voltage.

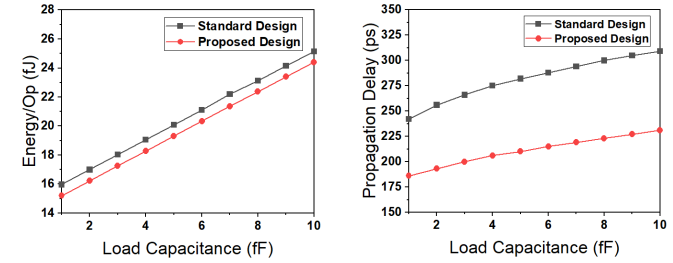


Fig. 5. Energy consumption and propagation delay of the proposed and standard designs across the load capacitance.

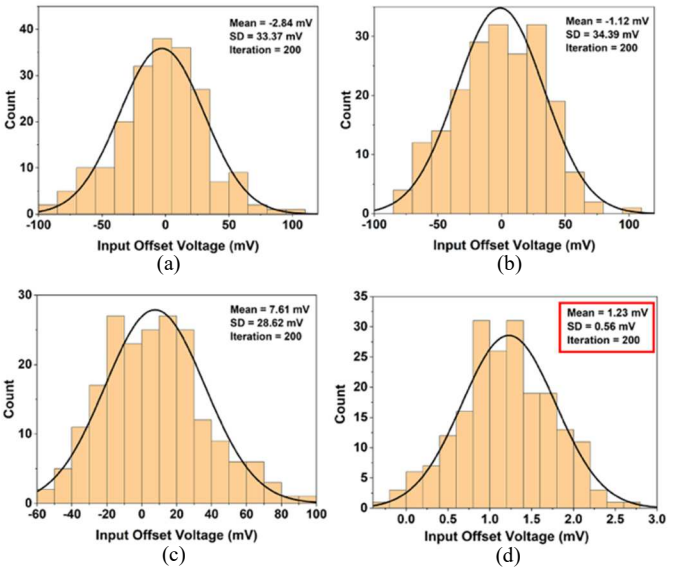


Fig. 6. Monte Carlo simulation results for the offset voltage of the proposed design (d) compared with the previous designs (a, b, and c).

means they are in the fast-fast condition. Fortunately, the proposed design is working well at different ranges of variations which makes it reliable to be used for industrial applications.

F. Monte Carlo Simulation

The energy consumption and propagation delay of the proposed design was tested for mismatch and process variations in a Monte Carlo simulation. There were a thousand iterations for each circuit. The mean and standard deviation results of the energy and delay were recorded. The normal distributions with sigma = 3 were plotted as shown in Fig. 8. The proposed design energy consumption mean is 19.32 fJ and 19.34 fJ, and the standard deviation is 0.03 fJ and 0.19 fJ for the mismatch and

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TABLE I. PERFORMANCE SUMMARY AND COMPARISON

Parameter	ISSCC'2009 [9]	Electronics Letters'2015 [10]	IEEE Access'2019 [13]	TCAS II'2020 [14]	**This Work**
Simulation Type	Post-Layout	Pre-Layout	Post-Layout	Post-Layout	**Post-Layout**
Technology (nm)	65	130	180	65	**65**
Supply Voltage (V)	1	1.2	1.2	1	**1**
Differential Input Voltage (mV)	18.6	NA	50	2	**10**
Maximum Frequency (GHz)	7	1.25	2	3	**8**[†]
Energy per Operation (fJ)	186	480	149	108	**19.31**
Propagation Delay (ps)	220	69	258	217	**211**
Offset Voltage (mV)	NA	7.78	7.3	2.05	**0.56**
Area (μm^2)	320	NA	252	603	**280**
Load Capacitance (fF)	NA	20	20	5	**5**
FoM[*] (fJ·ps·V)	204.6[**]	257.7	280.6	48	**2.28**

* FoM = Energy per Operation [fJ] x Propagation Delay [ps] x Offset Voltage [mV]
** Assuming Offset Voltage = 5 mV
† At V_{DD}= 1.2V, Load Capacitance = 1 fF, ΔVin = 50 mV

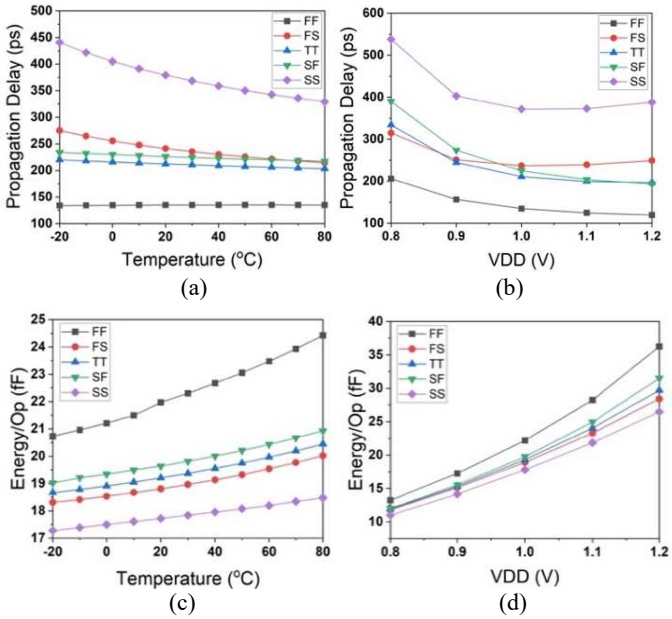


Fig. 7. PVT corners variation: (a) Propagation delay across temperature, (b) Propagation delay across voltage supply, (c) Energy consumption across Temperature, (d) Energy consumption across voltage supply.

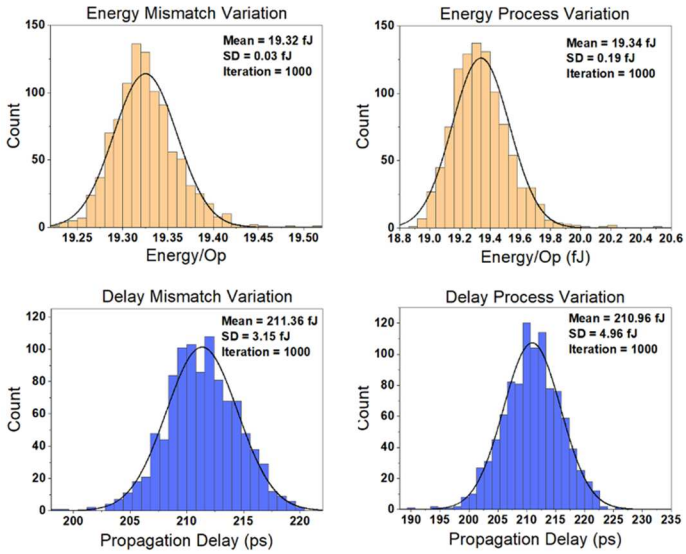


Fig. 8. Mismatch and Process Monte Carlo simulation results for the energy consumption and propagation delay of the proposed design.

process variation simulation, respectively. The propagation delay mean is 211.36 ps and 210.96 ps and the standard deviation is 3.15 ps and 4.96 ps for the mismatch and process variation simulation, respectively. Comparing the Monte Carlo simulation to the original simulation's energy consumption and propagation delay, there are similar trends when V_{diff} = 10 mV. Most importantly, the proposed designs experienced consistency when tested for process and mismatch variations.

The energy consumption and propagation delay are compared with the state-of-the-art comparator architectures as shown in Table I. It is noticed that the proposed design offers a low energy consumption among all other designs, and this improvement is caused by the new proposed architecture which disconnects the current path from the V_{DD} to the ground. The energy consumption is improved by 82% compared to the best energy consumption result in the table.

Moreover, the proposed design provides the best propagation delay and maximum frequency compared to other post-layout simulation results, and this improvement is initiated by the CFBB technique introduced in the proposed design. For the offset voltage, the proposed design shows a Monte Carlo result of 0.56 mV which is the best result compared to other designs.

VI. CONCLUSION

In this paper, the energy consumption, propagation delay and offset voltage have been improved by introducing a new schematic that turns the input transistors OFF in the reset phase and turns the input transistors ON in the evaluation phase. By doing this, the amplification gain in the evaluation phase is enhanced which results in a faster separation of output signals. Moreover, the transistors stack was reduced from 4 transistors to 3 transistors, which allows the design to be used at low voltage supply. The proposed design was post-layout simulated using TSMC 65 nm PDK and has been compared with other mainstream designs in the literature and showed an improvement in the energy consumption and propagation delay.

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